

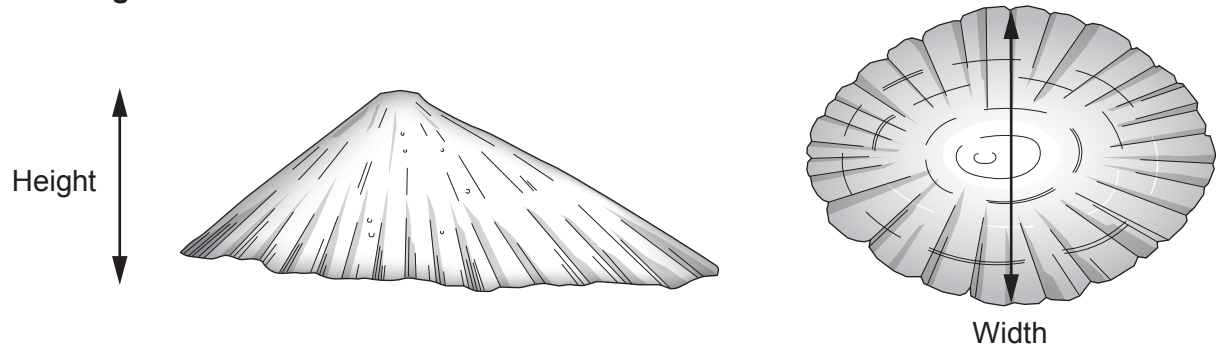
3. The common limpet (*Patella vulgata*) is found attached to rocks on most rocky shores in Wales. Limpets attach firmly to the rocks using a muscular foot. They also rotate their shell and grind it into rock to seal it into the rock. Limpets vary in height and width.



The common limpet

A survey was carried out to investigate the variation in the size of limpet shells on upper and lower shores in Cemlyn Bay, Anglesey in North Wales. The height and width of each shell were measured to the nearest mm as shown in **Image 3.1**.

Image 3.1



These measurements were used to calculate the height : width ratio.

$$\text{Ratio} = \frac{\text{height}}{\text{width}} : 1$$

- (a) (i) State what is meant by the term variation. [2]

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- (ii) State the type of variation shown in the size of the limpets. [1]

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Table 3.2

Limpet number	Limpet shell height : width ratio	
	Upper shore	Lower shore
1	1.35 : 1	0.38 : 1
2	1.53 : 1	0.36 : 1
3	1.47 : 1	0.41 : 1
4	1.80 : 1	0.28 : 1
5	3.11 : 1	0.44 : 1
6	3.00 : 1	0.57 : 1
7	2.56 : 1	0.58 : 1
8	2.45 : 1	0.37 : 1
9	2.00 : 1	0.27 : 1
10	2.42 : 1	0.30 : 1
Mean	2.17 : 1	 : 1

(b) (i) Calculate the mean shell height : width ratio for limpets sampled on the lower shore. **Write your answer in Table 3.2.** [2]
Space for working

(ii) State the conclusion that can be made about the height to width ratios of limpets and their position on the shore. Suggest a reason for this. [2]

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(iii) State how the students could increase confidence in their results. [1]

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